Report on Trade Performance Analysis

1. Methodology

Data Collection and Preprocessing

- The dataset consists of trading transactions from multiple accounts.
- Missing values were handled by filling them with zeros , Mean and Mode.
- Features were engineered to compute key performance indicators such as:
  - **Cumulative Profit & Loss (PnL)**: Measures overall profit progression.
  - **Total Positions**: The count of executed trades per account.
  - **Win Positions**: Number of profitable trades.
  - **Win Rate**: Ratio of profitable trades to total trades.
  - **ROI (Return on Investment)**: Measures profitability relative to investment.
  - **Maximum Drawdown (MDD)**: Indicates maximum loss from a peak.
  - **Sharpe Ratio**: Assesses risk-adjusted return using daily returns.

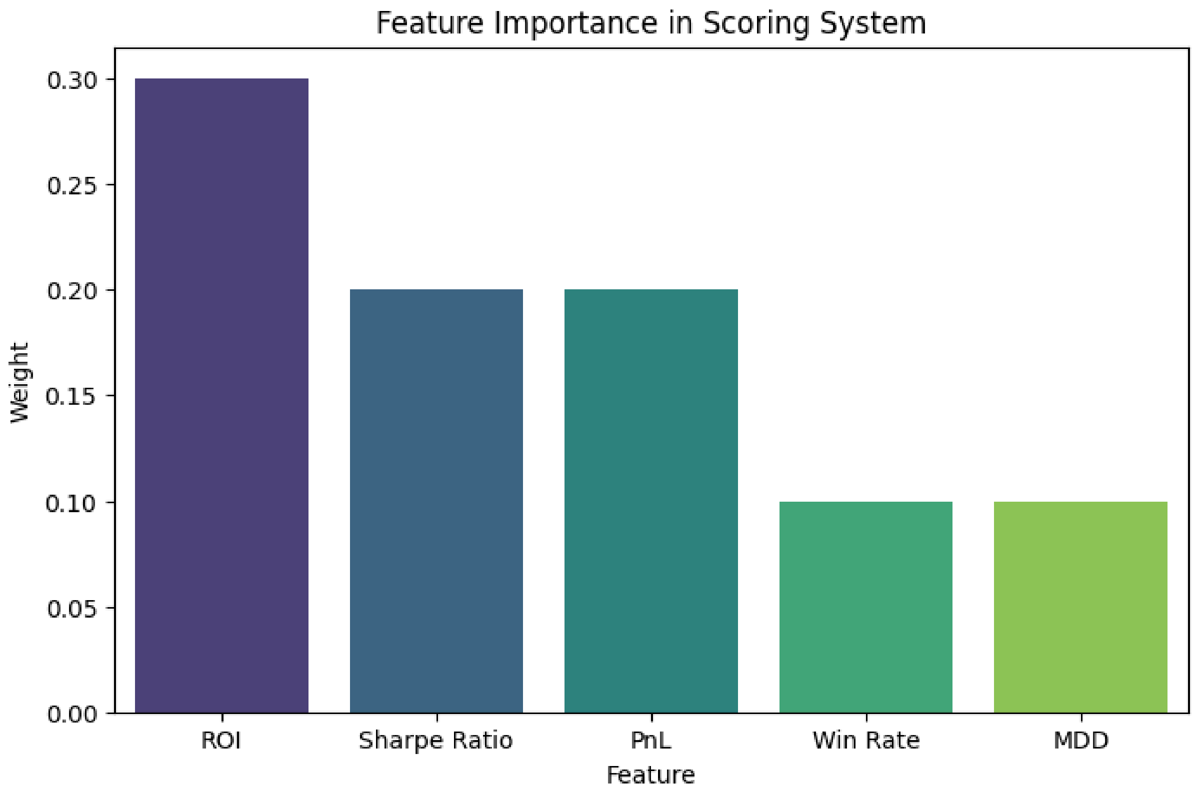
Feature Importance and Correlation Analysis

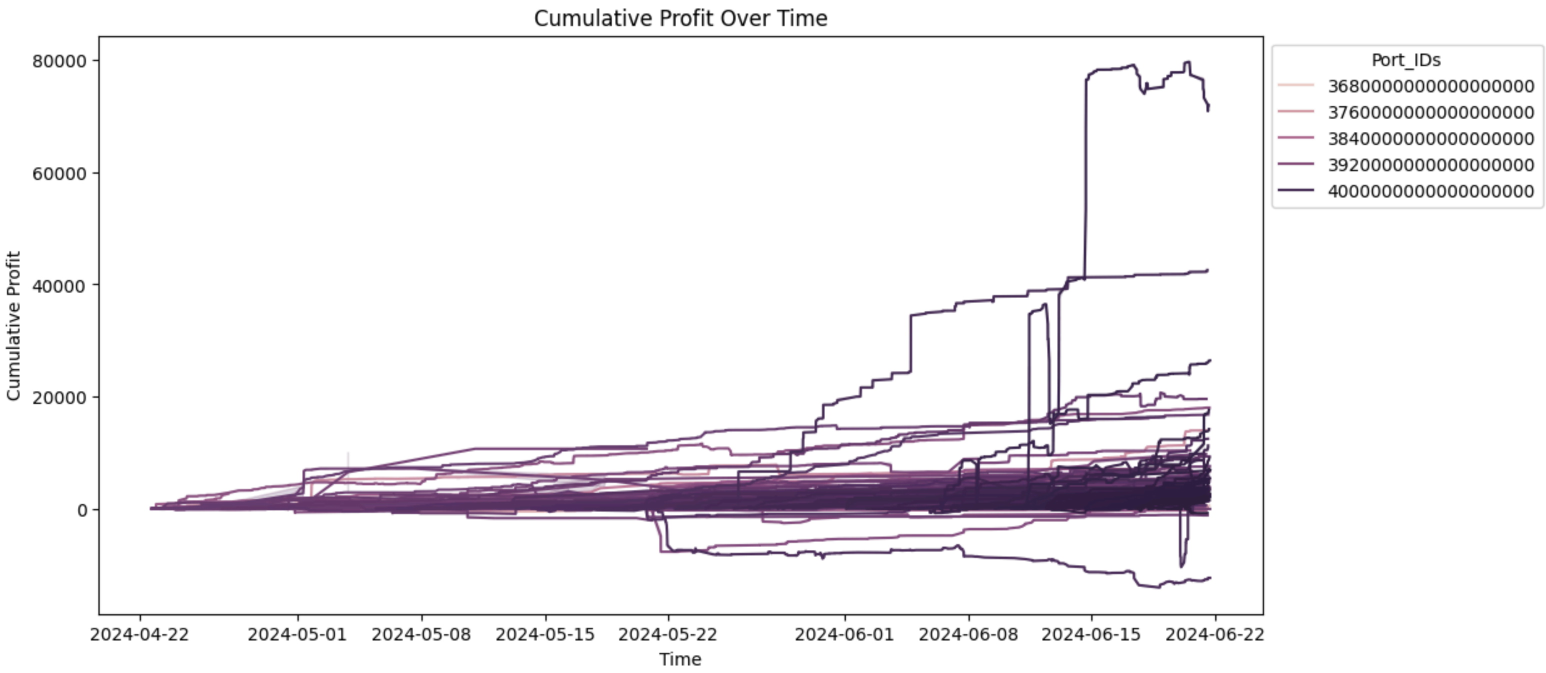
- A correlation matrix was generated to examine relationships between features.
- Strong correlations were observed between **Win Positions and Total Positions (0.95)** and **Sharpe Ratio and Win Rate (0.68)**.

- ROI had a moderate correlation with **Sharpe Ratio (0.49)**, suggesting its importance in risk-adjusted returns.

Scoring and Ranking Algorithm

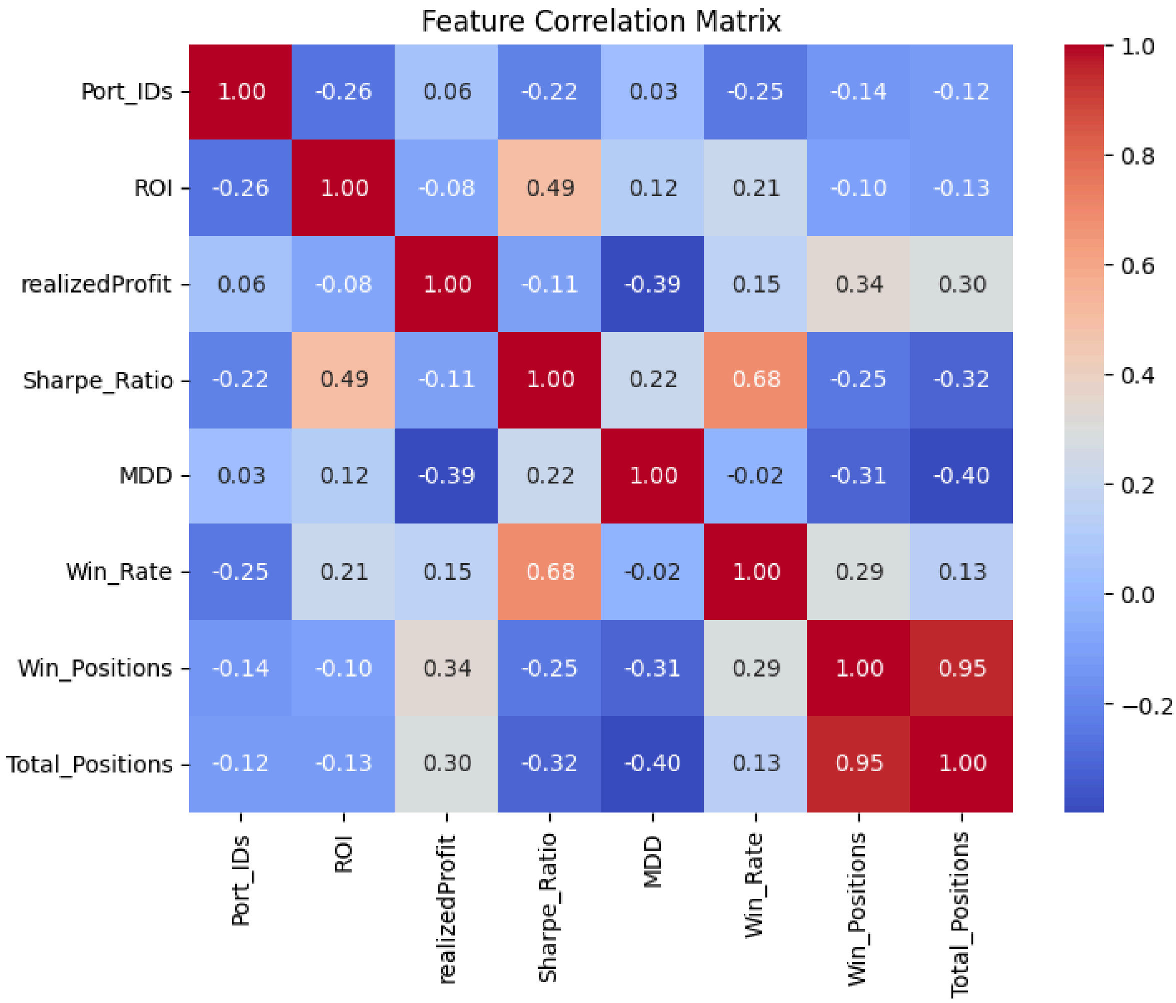
- A weighted scoring system was used to rank accounts based on the following factors:
  - **ROI (30%)**
  - **Sharpe Ratio (20%)**
  - **Realized Profit (20%)**
  - **Win Rate (10%)**
  - **MDD (10%)** (inverse score applied)
- The total weighted score was computed, and accounts were ranked in descending order.
- The top 20 accounts were selected based on their final scores.

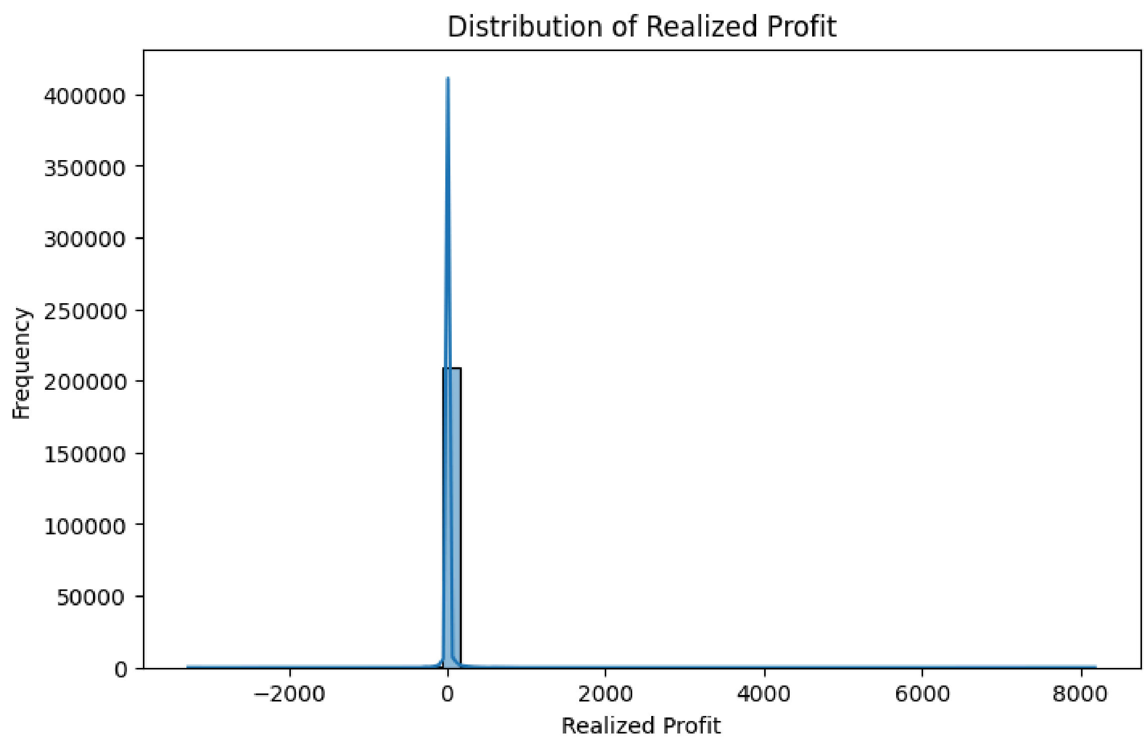




Visualization

- A **feature correlation matrix** was generated to highlight relationships between variables.
- A **cumulative profit over time** graph was plotted to analyze performance trends per account.





2. Findings

- The top-performing accounts exhibited **high Sharpe Ratios**, indicating strong risk-adjusted returns.

- **Win Positions and Total Positions showed high correlation**, suggesting that more active traders tend to have higher absolute profits.
- Accounts with **lower MDD** and **higher ROI** ranked higher due to their stability and profitability.
- The **distribution of ROI** was highly skewed, with a few accounts significantly outperforming others.

3. Assumptions and Limitations

- **Data Integrity**: Assumes the provided dataset is complete and accurate.
- **Stationarity**: Market conditions may change, affecting future ROI and Sharpe Ratios.
- **Risk Factors**: The analysis does not account for external market risks or leverage used in trades.
- **Scoring Weights**: The assigned weights were chosen based on general financial risk management principles and may not be optimal for all trading strategies.

4. Deliverables

- **CSV Files**:
  - calculated_metrics.csv: Includes all computed metrics for each account.
  - top_20_accounts.csv: Lists the highest-ranked 20 accounts.
  - summary_report.csv: Summary statistics of key features.
- **Visualizations**:
  - Correlation matrix

- Cumulative profit over time plot.

5. Conclusion

The analysis successfully identified the top 20 trading accounts based on performance metrics and risk-adjusted returns. Further refinements, such as dynamic weighting or integrating additional risk factors, could improve the robustness of the ranking system.